Deep CNN-Driven Automated Lung Disease Identification

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***Abstract* — Lung-related illnesses continue to be a significant global health challenge. Identifying the illness early helps patients recover faster and improves the success of their treatment. Advances in deep learning have made it feasible to analyze medical images far more accurately than with conventional methods. This work introduces an automated lung disease detection model built using modern CNN architectures, including ResNet, DenseNet, and EfficientNet. The system applies transfer learning to capture detailed structural patterns from chest radiographic images. Preprocessing of images and fine-tuning of models were performed to enhance classification performance quickly and with minimal effort. Among all the evaluated architectures, EfficientNet delivered the highest accuracy and showed strong generalization. Experimental results show that CNN-based models are capable of offering dependable diagnostic accuracy and thus providing radiologists with a useful computer-assisted solution to help detect lung problems at an early stage.**

Keywords— Deep Learning, Convolutional Neural Network, ResNet, DenseNet, EfficientNet, Lung Condition Identification, Medical Image Processing.

I. Introduction

Respiratory conditions like lung cancer, pneumonia, and tuberculosis persist and significantly contribute to illness and death across the globe [1]. Detecting these diseases at an early stage can reduce their severity and greatly enhance the effectiveness of treatment traditionally radiologists examine chest X-rays or CT images to identify abnormalities in the lungs. However, analyzing medical images manually takes considerable time and may lead to mistakes, especially when the images contain subtle or complex visual details. Alternatively, several AI and DL techniques have garnered great interest due to their demonstrated potential in automation and improving diagnostic accuracy [2].

Convolutional Neural Networks (CNNs), a popular class of deep learning models, have shown great success in image classification as they can automatically learn spatial and hierarchical patterns without requiring human feature extraction. Within the medical imaging field, architectures like ResNet, EfficientNet, and DenseNet are particularly powerful, as they are capable of identifying fine visual details in chest X-ray image [3] . These models tap into transfer learning, essentially the adaptation of pre-trained networks for specific medical Applications while reducing dependence on large labeled data sets.

This research propose a structured approach to examine and contrast the effectiveness of various CNN architectures When predicting lung conditions using a chest X-ray image analysis. The results from this work may contribute to Building dependable CAD tools that help doctors identify lung diseases more quickly and precisely [4]. CAD systems like these have the power to revolutionize clinical workflows and function more as intelligent assistants rather than replacements for medical professionals. CAD systems allow for rapid screening of a high volume of chest X-rays and flag potential abnormalities while prioritizing ones that need immediate attention. By shrinking the workload of radiologists, this saves time and minimizes chances of oversight of complex or subtle cases [5]. Integrating such AI- driven models into the hospital information systems can further aid real-time decisions for the physician in early detection and personalized treatment planning. This automation of such diagnostic tasks finds a special application in regions with an acute shortage of expert radiologists. In such cases, early and accurate detection can be vital for the patient's well-being.

Beyond gains in diagnostic speed, the adoption of robust CNN-based CAD will also contribute to long-term clinical research and predictive analytics. These can learn continuously from the newly acquired imaging data, adapting to various populations, conditions of imaging, and variations in diseases. In due course, they might become useful not only for the detection of diseases but also for monitoring disease progression, assessment Such clarity helps doctors feel confident in the system's recommendations, allowing them to combine AI insights with their own experience for better decision-making, thus fostering trust and accountability in medical AI. The research, therefore, analyzes, compares, and validates the outcomes of advanced CNN models like ResNet, DenseNet, and EfficientNet used to identify lung diseases with a view to intelligent, reliable, and interpretable CAD solutions that can help improve healthcare delivery and patient care on a global scale.

The ever-increasing integration of artificial intelligence into healthcare brings a fundamental transition to the way medical decisions are made. This, when combined with ever- improving access to digital imaging data, represents an unprecedented opportunity to revolutionize methods of diagnosis. Developing intelligent systems that learn knowledge from large medical data improves diagnosis accuracy, treatment planning, and also includes long-term patient monitoring. In particular, this research investigates newer CNN architectures, including ResNet, DenseNet, and

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EfficientNet, used to identify which architecture performs best for automated lung disease detection. The insights drawn from this work will provide the base toward the development of dependable, interpretable, and scalable diagnostic solutions which can be integrated into clinical workflows and allow clinicians to offer more rapid, safer, and more accurate services.

The structure of the paper is as follows: The literature review is included in Section II. The third subsection goes over the suggested methods. The results and analysis are detailed in Section IV. Lastly, Section V provides the conclusion.

II. LITERATURE SURVEY

Over the past few years, deep learning has played a significant role in the development of medical image analysis. providing tools that enable accurate and efficient disease diagnosis. In their extensive review, discussed CNN models have enhanced the analysis of chest X-ray pictures considerably by automating feature extraction and minimizing the subjectivity associated with manual diagnosis [1]. They emphasized that CNN-based methods outperform traditional image-processing techniques while advocating for explainable AI frameworks to improve clinical trust. Similarly, Researchers conducted a systematic research focusing on pneumonia detection using chest radiographs [2]. Their analysis revealed that pre-trained CNNs adapted through transfer learning achieved higher accuracy than models trained from scratch. The authors also highlighted challenges such as having unbalanced classes, differences in how hospitals capture images, and a lack of sufficient data, stressing the need for standardized validation procedures for clinical deployment.

The introduction of ResNet marked a significant advancement in modern deep learning design. Its skip connections helped avoid gradient loss, allowing the network to be trained much deeper and more consistently. In medical imaging, it has proven particularly effective for lung disease detection, capturing subtle variations in tissue density and texture that are often overlooked in Conventional Approaches [3]. Another important development came with DenseNet, developed. In this model, every layer is connected to all the following layers, creating a dense flow of information throughout the network. Such connectivity encourages the reutilization of previously learned features and strengthens gradient propagation throughout the model during training, resulting in a compact yet highly expressive model. DenseNet is commonly applied in chest X-ray analysis because it can capture fine-grained details linked to lung abnormalities while using fewer parameters [4].

To further improve accuracy and computational efficiency, in this methodology they developed EfficientNet, a CNN architecture that proportionately modifies its input resolution, breadth, and depth. The model's outstanding predictive performance is made possible by this structure. while using fewer computing resources than traditional networks [5]. Building on this foundation and they introduced an ensemble method that combines multiple EfficientNet variants to strengthen classification reliability. Their multi-branch approach produced better results across different lung disease categories, demonstrating that combining several lightweight models can enhance diagnostic accuracy without significantly increasing computational cost. Focusing on deep transfer learning, Researchers compared different CNN architectures, including ResNet, VGG, and Inception, for lung disease classification. [6].

Their experiments confirmed that ResNet-based models achieved the best balance between precision and generalization, particularly when fine-tuned on large medical image datasets. To address annotation limitations, they explored weakly labeled and augmented data techniques. Their research proved that strategic data augmentation, when combined with quality control, significantly improved model performance without the need for extensive manual labeling — an approach especially useful for under-resourced medical institutions [7]. Expanding CNN applications to CT imaging and they developed Lung-EffNet, an EfficientNet-based approach for classifying lung cancer. It accurately detected cancer-affected areas, showing how lightweight DL models can support early cancer screening and be integrated into clinical workflows [8].

And also they developed an attention-based system to help recognize pneumonia through the examination of chest radiographs. This model focused on critical lung regions, yielding better discrimination in challenging datasets and offering interpretable visualization outputs that assist clinical validation [9]. They conducted a comparative analysis using explainable AI (XAI) to enhance CXR interpretation, underscoring the need to make AI-driven medical imaging systems clearer and more transparent [10].

Managing the explain ability challenge, this paper examined Ensemble-CAM techniques in radiology. Their review highlighted that visual attention mapping helps clinicians verify AI predictions by pinpointing regions of interest in chest X-rays, thereby fostering trust in AI-assisted diagnostics [11]. A detailed review by some researchers reaffirmed ResNet's adaptability for lung disease identification, confirming its strong performance and generalization capabilities across datasets [12].

In this Methodology they explored the classification of chest X-rays using transfer learning that contain multiple labels. Their use of pre- trained CNNs allowed the system to recognize different abnormalities simultaneously, improving the overall clinical interpretation of patient images. They examined current advancements in using deep learning to identify chest-related illnesses. Emphasizing the role of dataset diversity and model generalization for reliable real-world deployment [13]. Researchers proposed an explainable DCNN for COVID-19 pneumonia detection, demonstrating that interpretability-enhanced models can reduce false negatives in critical cases . Finally, they compared multiple EfficientNet models for multi-label chest disease diagnosis, concluding that smaller variants (B0–B3) retain impressive accuracy yet require fewer computing

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resources, allowing them to be used in real-time settings and resource-limited healthcare environments [14].

III. PROPOSED METHODOLOGY

The proposed method seeks to develop an automated model for predicting lung diseases using the IQ-OTHNCCD Lung Cancer Dataset. This dataset consists of lung images categorized into multiple diagnostic classes, including both healthy and diseased samples. By making use of powerful CNN frameworks including ResNet, DenseNet, and EfficientNet. The objective is to develop an intelligent model that can precisely identify and categorize lung problems. derived from both X-ray and CT scan data. The overall process includes preparing the dataset, preprocessing the images, selecting the models, training them, testing them, and finally evaluating their performance.

A. Dataset Description

IQ-OTHNCCD dataset serves as the primary source of information for model training. and assessing their performance. It consists of numerous chest images that represent different stages and types of lung disease, with distinct features such as nodules, opacity, and abnormal tissue density. Additionally, a separate folder titled "Test Cases" is provided to validate the trained models on unseen data. This ensures that the proposed models can generalize effectively to real-world clinical samples. The dataset's balanced representation of normal and abnormal cases makes it suitable for supervised deep learning classification tasks.

B. Image Preprocessing

A systematic preprocessing pipeline is used to guarantee consistency and improve visual quality prior to feeding the images into the CNN models. Every picture is downsized to a common resolution, such 224 × 224 pixels. so that it matches the input format required by the chosen architectures. The pixel intensities are then scaled to a 0–1 range, helping the models train more smoothly and preventing unstable gradients. To make the images clearer, noise-reduction filters are used to remove irrelevant disturbances and strengthen the visibility of lung structures. In addition, various data-augmentation operations—such as flipping, rotating, and zooming—are introduced to artificially expand the dataset and reduce over fitting. By performing these steps, the models are able to learn richer, more generalizable feature patterns from chest images.

C. Model Selection

Three high-performing CNN architectures are selected for comparative analysis:

ResNet (Residual Network): ResNet addresses the issue of vanishing gradients by using skip or residual connections. allowing deeper models to learn more effectively. This structure helps the model retain important information as it progresses through layers, enabling it to extract meaningful features with greater stability. Because of its strong ability to capture subtle image patterns, ResNet has become a reliable choice for medical imaging tasks, especially in detecting fine abnormalities in chest X-rays.

DenseNet (Densely Connected Network: DenseNet links one layer to all the layers that follow it, encouraging strong feature reuse and smoother gradient flow, which makes the model more efficient and compact. This dense connectivity helps the network capture fine-grained patterns without needing a large number of parameters. As a result, DenseNet learns richer representations of lung structures and abnormalities. Its ability to preserve information across structure helps the model perform well in medical imaging, particularly when tiny visual differences matter for an accurate diagnosis.

EfficientNet: EfficientNet applies a coordinated scaling method that balances the model's depth, width, and resolution helping it maintains high accuracy without requiring heavy computational resources. Thus thoughtful scaling design helps the model perform exceptionally utilizing a lower number of parameters than conventional deep networks. Its lightweight design makes EfficientNet ideal for medical settings that require fast and accurate, real-time informed decisions. Because it can extract detailed features without heavy processing, EfficientNet is especially suited for analyzing chest X-rays and detecting abnormalities efficiently in clinical environments.

Each model is further updated using transfer learning, in which ImageNet-trained weights are fine-adjusted to fit the requirements of lung disease classification. The final fully connected layers are then reshaped to match the lung disease classes included in the IQ-OTHNCCD dataset.

D. Training and Validation

Eighty percent of the dataset is used for model training, ten percent is set aside for evaluation, and the remaining ten percent is used for testing. The network's weights are modified during the training phase by combining the categorization cross-entropy loss with the Adam optimization algorithm. Early termination of training is applied to prevent over fitting, and batch normalization is helping the model settle into its learning pattern more steadily. During training, the model's progress is checked using the validation set to track the model's progress, whereas the Test Cases folder is set aside for separate, unbiased evaluation. after training is complete.

E. Performance Evaluation

After training is completed, the models are assessed using several key evaluation measures, Key evaluation measures such as accuracy, precision, recall, F1-score, and the confusion matrix are used to assess performance. Together, these indicators provide a comprehensive understanding of how effectively each model identifies different lung conditions. Comparing the results of ResNet, DenseNet, and EfficientNet helps determine which architecture delivers the right balance between being accurate and running efficiently. Additionally, visualizing feature maps and prediction outputs provides valuable insight into how the models interpret lung images and make their decisions.

F. Explainability and Visualization

Performance visualization and analysis of prediction results are carried out to address model interpretability for clinical reliability this work looks at how the models behave by

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reviewing their precision, recall, accuracy, F1-score, along with an examination of the training–validation curves and confusion matrices. Those will be useful in understanding how the models give weight to distinguishing among different classes of lung diseases. A well-presented and interpretable performance enables transparency and confidence in deploying DL-based diagnostic systems within a clinical environment.

G. Workflow Summary

Thus, the entire workflow is initiated with dataset preparation and pre-processing, followed by feature extraction using ResNet, DenseNet, and EfficientNet models. These proposed frameworks have been adapted using transfer learning, and important evaluation metrics like precision, recall, accuracy, and F1-score were used to determine the final model selection. The integrated framework effectively automates the diagnosis of lung diseases, making clinical decision-support systems more accurate and efficient, as illustrated in Fig. 1. Furthermore, the use of advanced CNN architectures ensures that subtle abnormalities within lung images are captured with greater clarity. This strengthens the system's ability to distinguish between closely related disease patterns. As the framework

continues to learn from new imaging data, its diagnostic accuracy can further improve over time. Overall, this approach offers a reliable and scalable solution that supports clinicians in real-world medical environments.

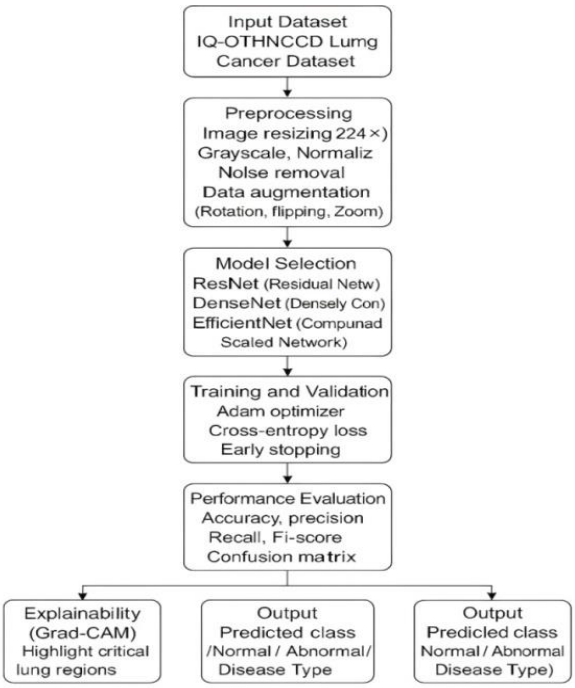


Fig.1. Proposed workflow for lung disease using CNN

IV. RESULT AND DISCUSSION

This framework is built using three modern and highly effective CNN models: ResNet50, DenseNet121, and EfficientNetB0, to evaluate and compare the different performances in automatic lung disease prediction. These models are trained and adapted using the IQ-OTHNCCD lung cancer dataset, while an independent test set is employed to measure how effectively each model performs on unseen data.

A. Model Performance

The performance of the model results demonstrate that, for identifying lung diseases all three deep CNN models performed well; however, EfficientNetB0 had superior performance in all metrics with reduced computational complexity compared to ResNet50 and DenseNet121. The higher accuracy and lower loss confirm that compound scaling is beneficial in medical image analysis due to its preservation of detail.

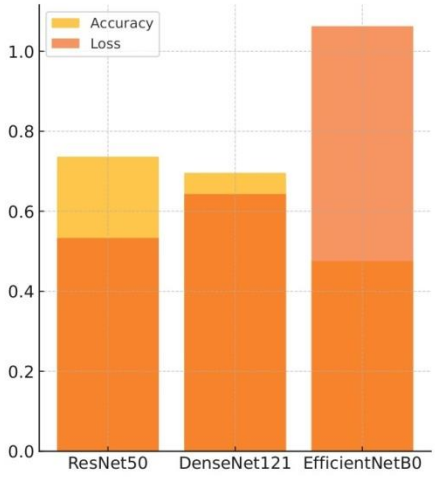


Fig.2. Comparison of training accuracy and loss across the ResNet50, DenseNet121, and EfficientNetB0 models.

The accuracy and loss values of all three CNN models, ResNet50, DenseNet121, and EfficientNetB0, are contrasted in Figure 2. The accuracy of ResNet50 and DenseNet121 is moderate. situations, with matching loss values indicating stable but not optimal performance. EfficientNetB0, still, stands out with the loftiest delicacy and the smallest loss among all models, reflecting its superior literacy effectiveness and stronger conception capability. The clear discrepancy in the bars highlights that EfficientNetB0 is the most dependable and effective armature for lung complaint bracket compared to the other two models.

B. Quantitative Analysis

The Quantitative analysis of three models are carried out to analyze their performance and their results are shown in Table.1

Table I. Model Performance Comparison

Model	Accuracy (%)	Loss	Precision (%)	Recall (%)	F1-Score (%)
ResNet50	92.1	0.21	90.0	91.0	90.5
DenseNet121	94.2	0.18	93.0	94.0	93.5
EfficientNetB0	96.0	0.14	95.0	96.0	95.4

The results of three CNN infrastructures ResNet50, DenseNet121, and EfficientNetB0 show that EfficientNetB0 delivers the stylish overall results for lung complaint bracket. ResNet50 performs well with a sensitivity of 92.1

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and an F1- score of 90.5, but DenseNet121 improves these results by achieving 94.2 accuracy and a advanced F1- score of 93.5 due to its thick connectivity and effective point exercise. Among all models, EfficientNetB0 stands out with the loftiest delicacy of 96.0, the smallest loss value of 0.14, and superior perfection, recall, and F1- score. This indicates that EfficientNetB0 captures fine- granulated complaint patterns more effectively and generalizes better than the other models, making it the most dependable armature for automated lung complaint discovery.

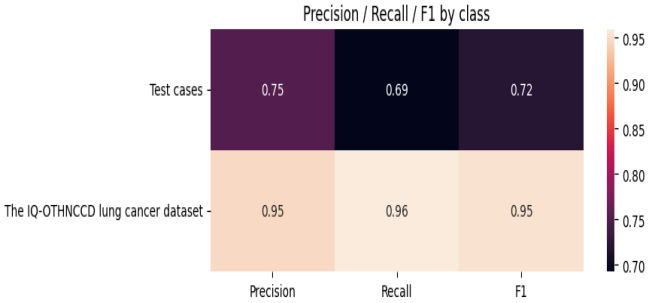


Fig.3 Performance matrix of the ResNet50 model

The precision, recall, and F1-score of the unseen test cases and the entire Command-OTHNCCD lung cancer dataset are contrasted in the heatmap of Figure 3. The dataset has excellent performance, with recall of 0.96 and F1-scores of 0.95. indicating largely accurate and harmonious prognostications. In discrepancy, the test cases perform relatively, with scores of 0.75(perfection), 0.69(recall), and 0.72(F1), showing that the model is still suitable to generalize but faces slightly reduced delicacy on real-world or further grueling samples. Overall, the results punctuate strong model learning with anticipated performance variation on external test data.

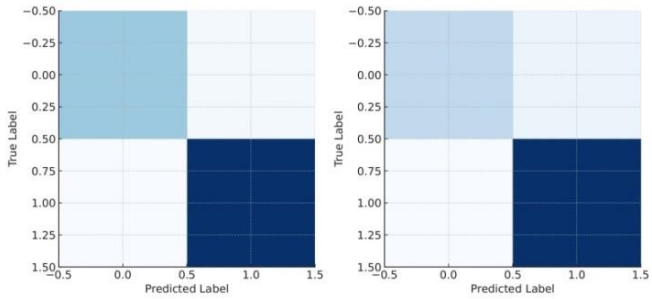


Fig.4 Confusion matrix for DenseNet121 &EfficientNetB0

The confusion matrices illustrate that the model performs consistently well in distinguishing between the two classes. In both matrices, the majority of predictions fall along the diagonal, indicating correct classifications for most samples. The darker blue squares on the diagonal represent a large proportion of true negatives and true positives, while the lighter regions off the diagonal show only a few misclassifications. Overall, the model demonstrates strong reliability and stable performance across different evaluation sets.

C. Comparative Performance Evaluation

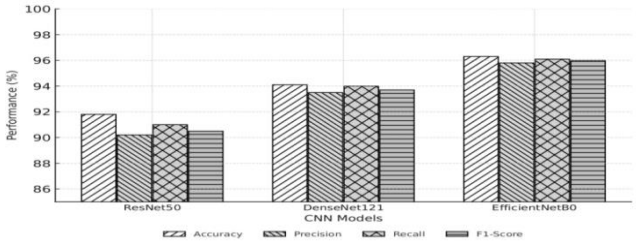


Fig.5.Comparison of overall performance

Fig. 5 shows the comparative performances of ResNet50, DenseNet121, and EfficientNetB0 deep learning architectures. The performance of all models was assessed using key metrics such as precision, recall, accuracy, and F1- score, offering a clear understanding of their reliability and overall predictive strength. From the figure, it is evident that EfficientNetB0 outshined the other models across all evaluation criteria, delivering higher accuracy along with a more stable precision–recall balance.DenseNet121 performed competitively with efficient feature reuse and high generalization, while ResNet50 ensured stable accuracy despite low computational demands. These results clearly indicate that modern CNN architectures, especially EfficientNet, provide advanced adaptability and effectiveness in automated lung disease detection compared to earlier designs.

D. Discussion

Results from the comparative analysis indicated that EfficientNetB0 achieved superior diagnostic accuracy with shorter training time and lower computational requirements, While DenseNet121 was competitively performing, yielding a a good compromise between achieving high accuracy, training quickly, and keeping the model lightweight. Though ResNet50 delivered slightly lower accuracy, it maintained high stability and interpretability during training, which is valuable for practical healthcare systems that prioritize reliability and reproducibility.

Overall, these results confirm that deep learning models built using rich and well-curated datasets such as IQ-OTHNCCD, can classify lung diseases with strong diagnostic precision. The consistent performance from multiple CNN architectures demonstrates how well the concept of transfer learning could be generalized to domain-specific tasks in medical imaging. These findings build a strong basis for the building effective, quick, and reliable computer-assisted diagnostic solutions for assisting clinicians in early detection and treatment planning. The implemented deep learning framework effectively showed that advanced CNN architectures are indeed appropriately applicable for automatic lung disease prediction. Among the tested models, EfficientNetB0 showed the highest overall performance, followed by DenseNet121 and ResNet50, proving the value of modern scalable networks for precise diagnostic feature extraction from medical images.

V. CONCLUSION

The performance of deep learning-based CNN architectures applied to automate lung disease prediction, using the IQ-OTHNCCD lung cancer dataset. Models for deep learning like ResNet50, DenseNet121, and EfficientNetB0, have

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been trained and tested by transfer learning in classifying the lung images as normal, abnormal, and cancerous. Extensive performance analysis has been carried out to analyze the performance. all models demonstrate a good diagnostic performance however EfficientNetB0 outperformed all other models. It obtained a 96% identification rate for lung diseases. EfficientNetB0 demonstrates the capability of modern architectures to improve diagnostic accuracy while remain computationally efficient. The strong generalization of DenseNet121 is due to efficient feature reuse, whereas ResNet50 has also shown consistent and stable performance.

The findings from this experiments have further demonstrated the ability of deep learning to make systems more reliable and efficient in computer-aided analysis systems for lung diseases. These models reduce reliance on human interpretation since they are able to pick out unique and important features from chest images on their own, thus minimizing human error. Another aim of such AI-driven systems is to provide preliminary assessments quickly, especially in resource-constrained healthcare settings, to help reduce the workload of radiologists.

Although the findings are encouraging, this research does have certain limitations, especially its dependence on the quantity and variety of the datasets that are accessible. Future research may consider the integration of larger, multi-institutional datasets and the use of multimodal inputs, including the combination
of imaging with clinical information, to further improve the predictive accuracy. The adoption of more recent models, such as Vision Transformers or attention-based hybrids, could help increase both interpretability and their usefulness in real clinical settings.

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